Team Solutions

1. The minute hand moves at a rate of 360 degrees per hour (or 6 degrees per minute), the hour hand moves at a rate of 30 degrees per hour (or 0.5 degrees per minute). Therefore, the angle between the hour and the minute hand increases by 5.5 degrees per minute. The first time the hands are perpendicular, the angle between them is 90 degrees. The second time is when the angle is 270 degrees, which occurs 180/5.5 or $\boxed{360/11}$ minutes later.

2. The product of the roots is 8, and the possible sets of distinct integer roots are $\{\pm1, \pm2, \pm4\}$, $\{\pm2, \pm2, \pm2\}$, and $\{\pm1, \pm1, \pm8\}$ where either zero or two of the roots are negative. Of all of the cases listed here, only one has distinct roots in arithmetic progression with two roots negative, $\{-4, -1, 2\}$. The corresponding polynomial is $(x-2)(x+1)(x+4) = x^3+3x^2-6x-8$, and the answer is $q = \boxed{3}$.

3. $n = 77k+24$ for some integer value of k. When n is divided by 7, the $77k$ term divides evenly, so the remainder is the residue of 24 modulo 7, or 3. Similarly when n is divided by 11, the remainder is 2. The answer is $3 + 2 = \boxed{5}$.

4. $\dfrac{1}{x+\sqrt{x}} + \dfrac{1}{x-\sqrt{x}} = \dfrac{2x}{x(x-1)}$. The domain of the function is all positive x except 1. For x between 0 and 1, the denominator is negative, so the function is negative. For x greater than 1, the function is strictly decreasing. It equals 1 when $\dfrac{2x}{x(x-1)} = 1 \rightarrow 2x = x^2 - x \rightarrow x^2 - 3x = 0 \rightarrow x = 3$, so the interval is $(1, 3)$, which has length $\boxed{2}$.

5. Considering that the non-zero digits of N look like binomial coefficients, N is the binomial expansion of $(1000 + 1)^4$. Since $1001 = 7 \times 11 \times 13$, $1{,}004{,}006{,}004{,}001 = 7^4 \times 11^4 \times 13^4$. The prime factorization of any factor of N will have 0 to 4 powers of each of the primes 7, 11, and 13, so the total number of factors is $5^3 = \boxed{125}$.

6. In order for $p(p(n)) = 6$, $p(n)$ must be 6, 16, or 32 (no other product of up to three digits k has the property that $p(k) = 6$). Treat the cases individually.

   $p(n) = 6$: If n has one digit, then $n = 6$. If n has two digits, then n is either 16, 23, 32, or 61. If n has three digits, the digits of n must be some permutation of 116 or 123. There are three of the former (ignoring duplicates) and six of the latter, so there are 14 numbers less than 1000 for which $p(n) = 6$.

   $p(n) = 16$: All digits will be powers of two. There are three two digit numbers: 28, 44, and 82, and $\binom{6}{4} - 3 = 12$ three digit numbers (think of the digits as distinct boxes and the four powers of two as identical balls to be distributed in three buckets, but we need to throw out the three cases where all four balls go into one bucket) so there are 15 numbers less than 1000 for which $p(n) = 16$.

   $p(n) = 32$: All digits will be powers of two. There are two two-digit numbers: 48, and 84, and $\binom{7}{5} - 6 - 3 = 12$ three digit numbers (again, think of the digits as distinct boxes and the five powers of two as identical balls to be distributed, but we have to remove the six cases where

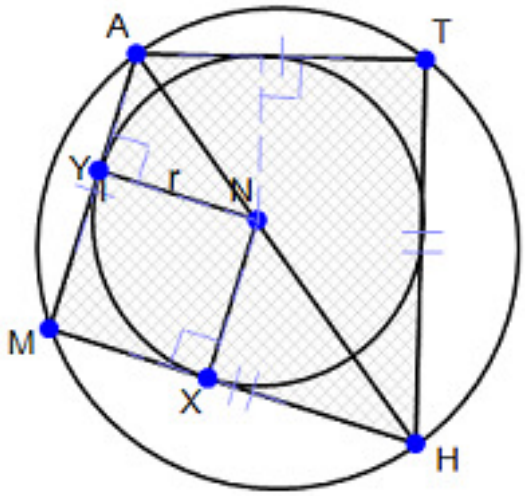
one box gets 4 balls and the three cases where one box gets 5 balls, so there are 14 numbers less than 1000 for which $p(n) = 32$.

The total is $14 + 15 + 14 = \boxed{43}$.

7. $x = 1 + \cfrac{1}{3+\cfrac{1}{1+\cfrac{1}{3+\cfrac{1}{\ddots}}}} \to x = 1 + \frac{1}{3+\frac{1}{x}} \to x - 1 = \frac{1}{3+\frac{1}{x}} \to 3x + 1 - 3 - \frac{1}{x} = 1 \to 3x^2 - 3x - 1 = 0 \to x = \frac{3 \pm \sqrt{21}}{6}$. The positive root is the value, so $A + B + C = 3 + 21 + 6 = \boxed{30}$.

8. Assume the center is blue. Then 1, 2, 3, or 4 of the corners must be blue. If 1 corner is blue, there are 4 possible corners to choose from and 2 ways to finish the grid with alternating red and green squares. If 2 corners are blue and the 2 corners are opposite, there are two ways to choose opposing corners and 2 ways to color each of the remaining paths of 3 squares with alternating colors. If 2 corners are blue and the corners are consecutive, there are 4 ways to choose consecutive corners, 2 ways to color the square between the two blue corners, and 2 ways to color the remaining squares. If three of the corners are blue, there are 4 ways to choose three blue corners, 2 choices for the color of each of the squares between 2 blue corners, and 2 ways to color the path of length 3. However, we must discard two of those cases that only contain 1 red or green square in the corner. If 4 corners are blue, there are $C(4, 2) = 6$ ways to pick the two green (or red) cells. Totaling these cases up we get $(4 \times 2) + (2 \times 2 \times 2) + (4 \times 2 \times 2) + (4 \times (2 \times 2 \times 2 - 2)) + 6 = 62$. Since there are three colors to choose for the center square, the answer is $\boxed{186}$.

9. Since AH is a diameter of length 20, we know that HAM is a right triangle with right angle at M. Let the center of the inscribed circle be N, and let radii of the inscribed circle intersect $\overline{MA}$ at Y and $\overline{MH}$ at X. Then $MYNX$ is a square of side length r. Further, triangles AYN and AMH are similar; thus $\frac{YN}{MH} = \frac{AY}{AM} \to \frac{r}{16} = \frac{12 - r}{12} \to 12r = 192 - 16r \to r = \frac{192}{28} = \boxed{\frac{48}{7}}$.



10. We can factor $x^3+y^3=(x+y)(x^2-xy+y^2)$, so $517832=98(x^2-xy+y^2) \rightarrow x^2-xy+y^2=5284$. In addition, $(x+y)^2=x^2+2xy+y^2=98^2=9604$. Subtracting the first equation from the second, we get that $3xy=4320$, so the area of the triangle is $\frac{xy}{2}=\frac{4320}{6}=\boxed{720}$.

11. Let $n=10+t$, then the expression is $(10+t)^2+(10+t)-9=t^2+21t+101$. For this value to be a multiple of 101, it means that $t^2+21t=t(t+21)$ must be a multiple of 101. Since 101 is prime, one of the two factors must be a multiple of 101. Since t can be at most 89, $t+21=101 \rightarrow t=80 \rightarrow n=\boxed{90}$.

12. A and B must be on different branches of the absolute value graph, and the x-coordinates of O, A, and B must be evenly spaced. Let the points be $O(0,0)$, $A(t,4-t)$, and $B(2t,2t-4)$. Equating OA and AB, we are looking for the solution to $\sqrt{t^2+(4-t)^2}=\sqrt{t^2+(3t-8)^2}$. Squaring both sides and subtracting t^2 from each side yields $(4-t)^2=(3t-8)^2$, yielding two solutions, $t=2,3$. If $t=2$, the points are $(0,0),(2,2)$, and $(4,0)$, which are not collinear. If $t=3$, the points are $(0,0),(3,1)$, and $(6,2)$, which are collinear, so $k=\boxed{1/3}$.

13. Assume the circle has center (x,y) and radius r. The following three equations must be satisfied:

$$\begin{aligned}(x-9)^2+y^2 &= r^2\\ (x-8)^2+(y-7)^2 &= r^2\\ x &= r.\end{aligned}$$

    Subtracting the first equation from the second, we get $2x-17-14y+49=0 \rightarrow 14y-2x-32=0 \rightarrow x=7y-16$. Squaring the third equation and subtracting it from the second, we get $-16x+64+(y-7)^2=0$. Substituting $x=7y-16$ gives $-16(7y-16)+64+(y-7)^2=0 \rightarrow y^2-126y+369=0$. Therefore, y is either 3 or 123, with corresponding values of x (and radii) of 5 and 845. The sum of the radii is $\boxed{850}$.

14. Let $x=\underline{G}\,\underline{E}\,\underline{R}$, $y=\underline{M}\,\underline{A}\,\underline{N}$, then $2(1000x+y)=5(1000y+x) \rightarrow 1995x=4998y \rightarrow \frac{x}{y}=\frac{4998}{1995}=\frac{238}{95}$. Since both x and y must be three digit integers with all non-zero digits, we multiply the numerator and denominator by 3 to get $x=714$ and $y=285$, rearranging the digits to get $\underline{E}\,\underline{N}\,\underline{G}\,\underline{R}\,\underline{A}\,\underline{M}=\boxed{157482}$.

15. It is fair to assume that each face of a polyhedron with the maximal number of interior diagonals is a triangle (otherwise some diagonals are on the faces of the polyhedron). If a triangular faced polyhedron P has F faces, E edges, and V vertices, then it has $\binom{V}{2}-E$ interior diagonals. For a polyhedron with all triangular faces, $3F=2E$. Using Euler's formula for polyhedra ($V+F=E+2$), we get $V+\frac{2E}{3}=E+2 \rightarrow E=3V-6$, so the maximal number of interior diagonals is $\binom{V}{2}-3(V-2)=\frac{(V-3)(V-4)}{2}$, which for $V=12$ is $\boxed{36}$. A sample polyhedron that has 36 interior diagonals is two convex pyramids that share a common decagon base.